

DPEN021: Enabling Mathematics A
Class Questions Week 1

Week 1A

Key Idea – Substitution

1. Find the value of each of the following expressions if $a = 4$, $b = -3$, $c = 1/4$:

- | | | |
|------------------|------------------------------|---------------------------------|
| (a) $a + b$ | (b) $a - b$ | (c) $b - a$ |
| (d) $c \times a$ | (e) $(a + b) \times (a - b)$ | (f) abc^2 |
| (g) $a + c$ | (h) $\sqrt{a} + b$ | (i) $\frac{a}{b} - \frac{b}{c}$ |

Key Idea – Simplification

1. Simplify the following:

- | | |
|---|--|
| (a) $3a + 5c - 9a + 4b - a + 2b$ | (b) $4b^3 + c^3 - 2abc - a^3 - b^3 - \frac{1}{3}abc$ |
| (c) $4q^2 - 4r^2 - 2r + 12$ | (d) $a^3 - 3ab + b^2 - 2a^3 - a^2 + 3b^2 + 5ab$ |
| (e) $9a^2 + 6a - 4 - 5a^2 - 9b + 6$ | (f) $xy - 3x^2 - 4y^2 + 6xy - 2y^2$ |
| (g) $6 - 2x - 7x^2 - 12 + 5x^2 + 2x$ | (h) $2a^2b + 3ab^2 + 4a^2b - 7ab^2$ |
| (i) $a^4 + 3a^2 - 2a - 7a^2 + a^3 + 6a^2$ | (j) $15xy^2 - 8x^2y + 10xyz - 4x^2y + 5xyz$ |

2. Simplify the following:

- | | | |
|----------------------|----------------------------|-------------------------------------|
| (a) $5x^2 \times 2y$ | (b) $3a^2b^3 \times 2a^4b$ | (c) $(-2a) \times -5 \times (-3ab)$ |
| (d) $2x^2 \div 4x$ | (e) $8a^2b \div (-16ab^2)$ | (f) $(-20x) \div (-10x^3y)$ |

Key Idea – Expanding using Distributive Law

1. Expand and simplify the following:

- | | |
|------------------------------|---------------------------------|
| (a) $-3a(a - b + c)$ | (b) $m(2m - 7) - 5m(4m - 4)$ |
| (c) $4a - 2b - (7 + 3b - a)$ | (d) $-abc(abc - a^2bc - ab^2c)$ |
| (e) $4a - (a + b) - (a - b)$ | (f) $2y(6 + 4y) - 6y(2 - 3y)$ |

Week 1B

Key Idea – Expanding using Distributive Law Twice

1. Expand the following products:

- | | | |
|-------------------------|------------------------|------------------------|
| (a) $(x + 3)(x - 1)$ | (b) $(n + 2)(2n + 6)$ | (c) $(2m - n)(3m + n)$ |
| (d) $(2x^2 + x)(x + 4)$ | (e) $(3c + 4)(5c + 2)$ | (f) $(a - 3)(2a - 5)$ |
| (g) $(2x + 3y)(x + 2y)$ | (h) $(a + b)(3a - 4b)$ | (i) $(a - b)(3a - 4b)$ |

Key Idea – Expanding Binomial Products

1. Expand the following:

- | | | |
|---------------------|-------------------|------------------|
| (a) $(x + 3)^2$ | (b) $(a + 6)^2$ | (c) $(x - 4)^2$ |
| (d) $(3a - 1)^2$ | (e) $3(1 - 5y)^2$ | (f) $(1 - 5y)^2$ |
| (g) $(m^2 + n^2)^2$ | (h) $(4a - 3b)^2$ | (i) $(ab - 3)^2$ |

2. Expand the following resulting in a difference of two squares:

- | | | |
|--------------------|------------------------|--------------------------|
| (a) $(x-4)(x+4)$ | (b) $3(2m-7)(2m+7)$ | (c) $(2y+1)(2y-1)$ |
| (d) $(ab+1)(ab-1)$ | (e) $(5x+3)(5x-3)$ | (f) $(2m+n)(2m-n)$ |
| (g) $(a-7b)(a+7b)$ | (h) $(3x^3-y)(3x^3+y)$ | (i) $(x^2+y^2)(x^2-y^2)$ |

Key Idea – Further Expansions

1. Expand and simplify the following:

- | | | |
|------------------------|-------------------------|-------------------------------|
| (a) $(a+1)(a^2+3a+3)$ | (b) $(m-6)(2m^2+3m+1)$ | (c) $(3a^2+a)(a^3-3a^2+6a-3)$ |
| (d) $-3(a+3)-(2a-1)^2$ | (e) $(3-x)(2-x)(x+1)$ | (f) $(a+1)(a-3)(a+4)$ |
| (g) $(x-1)(x^2-2x+1)$ | (h) $(2a-1)(4a^2+2a-1)$ | (i) $5(3b-1)^2(b+2)$ |

Key Idea – Pascal Triangle

1. Write down the expansion for the following using Pascal's Triangle.

- | | | |
|----------------|----------------|----------------|
| (a) $(a+3)^3$ | (b) $(2+x)^3$ | (c) $(2a-b)^4$ |
| (d) $(2x+y)^4$ | (e) $(2x-1)^5$ | (f) $(2-x)^6$ |

Week 1C

Key Idea – Simple Common Factors

1. Factorise the following by taking out the HCF:

- | | | |
|-----------------------|---------------------|--------------------------|
| (a) $24-18y$ | (b) $7m+7n$ | (c) $4x+8$ |
| (d) $6a-18b$ | (e) $2y^2+4y$ | (f) at^2-2at |
| (g) $16m^2+12m$ | (h) $-4m+6n$ | (i) $7ab+4bc$ |
| (j) $3mn^3-9mn$ | (k) $5a^3-25a^2b$ | (l) $ax+ay+az$ |
| (m) $4xa^3+5x^2-2x$ | (n) $15x^2+5xy-25x$ | (o) $7ab-7bc+21b$ |
| (p) $m(x+y)^3+n(x+y)$ | (q) $2(y-1)-y(y-1)$ | (r) $(3+a)^7-(3+a)^{10}$ |

Key Idea – Compound Common Factors

1. Factorise the following by grouping in pairs:

- | | | |
|-----------------------|------------------------|----------------------|
| (a) $3a+3b+ax+bx$ | (b) $5x+5y+ax+ay$ | (c) $ab+3bc+4a+12c$ |
| (d) $9x^2+12x-3xy-4y$ | (e) $3ab-3ac+7b-7c$ | (f) $2a^3-a^2-10a+5$ |
| (g) $8b-80c-b^2+10bc$ | (h) $3x^2-9xy-xy+3y^2$ | (i) $x^5-x^3-x^2+1$ |
| (j) $4a-4b+ay-yb$ | (k) y^3-3y^2-y+3 | (l) $a^2-b^2+2a-2b$ |

Key Idea – Difference of two Squares

1. Use the difference of two squares method to factorise the following expressions:

- | | | |
|----------------|-----------------|-----------------|
| (a) a^2-b^2 | (b) x^2-4 | (c) $9-4b^2$ |
| (d) $25a^2-4$ | (e) $25-x^2$ | (f) x^2-1 |
| (g) $64-36m^2$ | (h) $4x^2-9y^2$ | (i) a^2-81b^2 |

- | | | |
|----------------------------|----------------------------|------------------------|
| (j) $24a^2 - 6b^2$ | (k) $1 - 64m^2$ | (l) $x^2 - 7$ |
| (m) $z^2 - \frac{1}{4}x^2$ | (n) $\pi R^2 - \pi r^2$ | (o) $49x^2 - 64y^2z^2$ |
| (p) $x^4 - 100$ | (q) $x^4 - y^4$ | (r) $(x+a)^2 - b^2$ |
| (s) $(x+y)^2 - (x-y)^2$ | (t) $(3x+4y)^2 - (2x-y)^2$ | (u) $21a^4 - 21b^4$ |

Key Idea – Sum and Differences of Two Cubes

1. Factorise the following by using the method of difference/sum of two cubes:

- | | | |
|-----------------------|--------------------------|-------------------------|
| (a) $x^3 + 27$ | (b) $x^3 - 64$ | (c) $t^3 - 1$ |
| (d) $x^3 + 125$ | (e) $y^3 - 1000$ | (f) $m^3 + 1$ |
| (g) $8a^3 + 1$ | (h) $3x^3 + 81$ | (i) $27a^3 - 1$ |
| (j) $8x^3 - 125$ | (k) $(2x+3)^3 - (x-4)^3$ | (l) $(x+5)^3 + (x-2)^3$ |
| (m) $p^7x^4 - p^4x^7$ | (n) $8 - (2-x)^3$ | (o) $2(x-y)^3 + 54$ |

Week 1D

Key Idea – Monic Quadratic Factorisation

1. Write down the factors of the following monic quadratic trinomials:

- | | | |
|----------------------|-----------------------|----------------------|
| (a) $x^2 + 14x + 45$ | (b) $x^2 - x - 42$ | (c) $x^2 - 15x + 36$ |
| (d) $x^2 + x - 12$ | (e) $x^2 - 10x + 24$ | (f) $x^2 - 12x + 36$ |
| (g) $x^2 - 10x + 25$ | (h) $x^2 + 2x + 1$ | (i) $x^2 - 4x + 4$ |
| (j) $x^2 - 12x + 35$ | (k) $x^2 - 29x + 100$ | (l) $x^2 - 2x - 35$ |
| (m) $x^2 - 14x - 15$ | (n) $x^2 - x - 42$ | (o) $x^2 - x - 110$ |

Key Idea – Non-Monic Quadratic Factorisation

1. Write down the factors of the following non-monic quadratic trinomials:

- | | | |
|---------------------------|-----------------------|-----------------------|
| (a) $3x^2 + 5x + 2$ | (b) $2x^2 + 5x + 2$ | (c) $2x^2 + 7x + 3$ |
| (d) $6x^2 + 23x + 7$ | (e) $2x^2 - 11x + 12$ | (f) $3x^2 - 10x + 8$ |
| (g) $2x^2 + 9x - 18$ | (h) $6x^2 - x - 12$ | (i) $2x^2 + 5x - 12$ |
| (j) $3x^2 + 19x - 14$ | (k) $2x^2 + 9x - 18$ | (l) $8x^2 + 22x + 15$ |
| (m) $3x^2 - 13x + 14$ | (n) $1 - 12x + 35x^2$ | (o) $12x^2 + 7x - 12$ |
| (p) $4x^2 + 16xy + 15y^2$ | (q) $5x^2 - 33x + 18$ | (r) $12 - 7x - 10x^2$ |

Key Idea – Combined Factorisation

1. Combined factorisation - fully factorise each of these expressions:

- | | | |
|-----------------------|------------------------|---------------------------------|
| (a) $3x^2 - 6xy$ | (b) $3x^2 - 6xy - 15$ | (c) $x^2y^2 - 100$ |
| (d) $-x^2 + 3x - 2$ | (e) $a^3bc - ab^3c$ | (f) $2x^3 + 14x^2 - 16x$ |
| (g) $a^3 - a$ | (h) $4x^2 - 28x - 480$ | (i) $p^2 - q^2 + 3p + 3q$ |
| (j) $16b^3 + 54$ | (k) $(x+y)^2 + 3(x+y)$ | (l) $4a^2y + 4a^2 + 4axy + 4ax$ |
| (m) $9x^2 - 39x + 30$ | (n) $-3a^2 + 6a + 24$ | |